

**Mid-Semester Exam**

**Questions:**

1. A parallelogram has vertices at  $A(2, -1, 4)$ ,  $B(1, 0, -1)$ ,  $C(1, 2, 3)$ , and  $D$ . Find
  - (a) the coordinates of  $D$ ,
  - (b) the cosine of the interior angle at  $B$ ,
  - (c) the vector projection of  $\overrightarrow{BA}$  onto  $\overrightarrow{BC}$ ,
  - (d) the area of the parallelogram,
  - (e) an equation for the plane of the parallelogram.

2. Let the linear change of variables be

$$u = 3x + 2y, \quad v = x + 4y$$

Let  $R$  be the region in the first quadrant bounded by the four lines

$$y = -\frac{3}{2}x + 1, \quad y = -\frac{3}{2}x + 3, \quad y = -\frac{1}{4}x, \quad y = -\frac{1}{4}x + 1.$$

- (a) Sketch  $R$  in the  $xy$ -plane and, under the map  $(x, y) \mapsto (u, v)$ , sketch its image  $S$  in the  $uv$ -plane.
- (b) Compute the Jacobian  $\frac{\partial(x, y)}{\partial(u, v)}$ .
- (c) Evaluate

$$\iint_R (3x^2 + 14xy + 8y^2) \, dx \, dy$$

by changing variables to  $(u, v)$  and integrating over  $S$ .

3. Find the extreme values of  $f(x, y, z) = x(y + z)$  on the curve of intersection of the right circular cylinder  $x^2 + y^2 = 1$  and the hyperbolic cylinder  $xz = 1$ .

Please note: There is no partial marking. Only integer marking. **All questions are worth 10 marks each.**